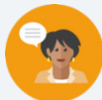


6.3 Writing and Solving Inequalities

Lesson Introduction

 Benchmark	MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.		 Learning Targets	• I can write and solve a two-step, one-variable inequality. • I can write and solve a multi-step, one-variable inequality. • I can write and solve a compound one-variable inequality. • I can interpret solutions to problems as viable or non-viable options.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.2.2 Solve two-step, linear inequalities in one variable, and represent solutions algebraically and graphically.		MA.912.AR.3.3 Given a mathematical or real-world context, write and solve one-variable, quadratic inequalities over the real number system. Represent solutions algebraically or graphically. MA.912.AR.4.2 Given a mathematical or real-world context, write and solve one-variable, absolute value inequalities. Represent solutions algebraically or graphically.	
 Essential Questions	• How can you write and solve a one-variable inequality representing a real-world context? • How can you determine if a solution is viable or non-viable in a real-world context?			
 Lesson Narrative	In this lesson, students extend their learning of inequalities from solving in mathematical contexts to writing inequalities that represent a real-world context, solving them, and interpreting them as viable or not. Students begin by collaborating to match given inequalities to real-world contexts. They will engage in an Exploration where they will look at context and viability and extend that concept to writing and solving inequalities to represent real-world contexts.			
 Component Summary	6.3.1 Warm-Up (~5 min)	Notice and Wonder activity (SE p. 305)	6.3.4 Guided Instruction (10 min)	Introduction to real-world inequalities (SE p. 307)
	6.3.2 Collaboration (10 min)	Match given inequalities to real-world contexts and solve (SE p. 305)	6.3.5 Activity (15 min)	Gallery Walk with real-world inequalities (SE p. 308)
	6.3.3 Exploration (5-10 min)	Real-world inequality Exploration activity (SE p. 306)	6.3.6 Wrap-Up (~5 min)	Growth Mindset self-reflection (SE p. 310)
 Resources	Glossary	Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A	• Act 1, 2, and 3 Videos for 6.3.3 Exploration.		• 6.3.5 Activity: Gallery Walk - students could receive the paper with contexts or have the scenarios hung around the room.

 Teacher Tips	Common Misconceptions • Students may forget to reverse the inequality symbol when multiplying or dividing by a negative. • Students may confuse when to use an open or closed circle when graphing the solution on the number line. • Students may confuse an “and” inequality with an “or” inequality when graphing. • Students may confuse writing the correct inequality when there are words such as “as much as” or “no more than”.	Things to Consider • For 6.3.5 Activity, the contexts will need to be prepared ahead of time to provide to students to complete the activity. Consider pacing when deciding whether to do all problems or a modified amount or process.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 6.3.1 Warm-Up provides students with an opportunity to review circles and x’s on a coordinate grid to determine what they notice about the y-values. This routine provides a low entry point for students regardless of their mathematical proficiency.	MA.K12.MTR.7.1: Real-World Contexts All the core components of this lesson provide students with multiple problems involving the real-world application of linear inequalities. During these activities students will connect this mathematical concept to everyday experiences and to see how it can be applied to other disciplines.
 Differentiate Instruction	Consider modifying activities to provide additional scaffolding if needed. For example, in 6.3.5 Activity, consider providing one of the answer choices for students who need the additional support and arrange the groups in such a way that only the students who need the additional support receive it.	
Additional differentiation opportunities are denoted throughout the lesson guide.		
Lesson Notes:		

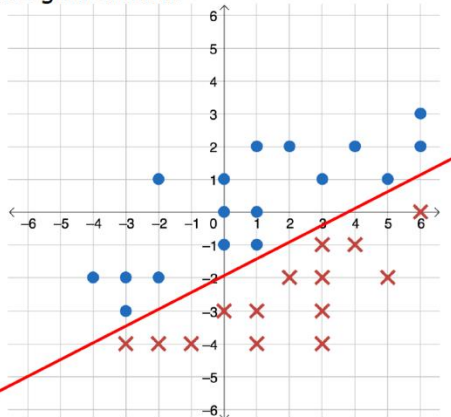
6.3.1 Warm-Up: Notice and Wonder

Component Overview: Students will complete a Notice and Wonder activity related to coordinates on a grid.



6.3.1 Warm-Up: Notice and Wonder

1. Draw a line that separates the circles and the Xs on the coordinate grid shown.



- a. Find any X whose x -coordinate is greater than the x -coordinate of a circle. Write their ordered pairs.

X	Circle
(3, -1)	(2, 2)

- b. What do you notice about the y -coordinates?
Answers will vary. The y -coordinate of the circle is greater than the y -coordinate of the X. The y -coordinate of the circle is positive and the y -coordinate of the X is negative.

- c. Find any X whose x -coordinate is the same as the x -coordinate of a circle. Write their ordered pairs.

X	Circle
(3, -1)	(3, 1)

- d. What do you notice about the y -coordinates?
Answers will vary. The y -coordinate of the circle is greater than the y -coordinate of the X. The y -coordinate of the circle is positive and y -coordinate of the X is negative.



This activity leverages the Notice and Wonder instructional routine. Using this routine makes this mathematical task more accessible for all students.

6.3.2 Collaboration: Inequalities in Real-World Contexts

Component Overview: Students will work with a partner to match each of the three inequalities to one of three real-world scenarios.



6.3.2 Collaboration: Inequalities in Real-World Contexts

Work with your partner to complete the following.

- Three inequalities are given.

$$5h + 4.5 \leq 20 \quad 4.5 \leq 5k \leq 20 \quad 5m > 20 + 4.5m$$

Match each of the inequalities to one of the situations below. Then, solve the inequality and explain how it matches the situation.

Situation A	
Markus went to his local coffee shop to get a latte and some coffee beans. The latte cost \$4.50 and beans cost \$5 per pound. He has a \$20 bill and wants to know how many pounds of beans he can buy.	
Matching Inequality and Solution	How does the inequality match the situation?
$5h + 4.5 \leq 20$ $5h \leq 15.5$ $h \leq 3.1$	Answers will vary. The variable is per pound, h, which is being multiplied by the \$5 cost per pound. Then that value is being added to the \$4.50 latte, which the total between the two purchases must be less than or equal to the \$20 that Markus has to spend.



The goal of this Collaboration activity is for students to be able to identify when a situation requires a compound inequality versus a single inequality, and when a situation would involve the variable on both sides.



How did you use the information given in the context to determine which inequality matched this situation?

6.3.2 Collaboration: Inequalities in Real-World Contexts (continued)

Situation B

At Rolling Lanes Bowling Alley, the cost is \$5 per game for non-members and \$4.50 per game for members. There is a \$20 annual fee to be a member. Luna wants to know how many games she needs to play in a year in order for the membership to be a good deal.

Matching Inequality and Solution

$$\begin{aligned} 5m &> 20 + 4.5m \\ 0.5m &> 20 \\ m &> 40 \end{aligned}$$

How does the inequality match the situation?

Answers will vary. Luna pays \$5 per game as a non-member. This inequality determines how many games that what she is currently spending is greater than the member rate of \$4.50 per game, m , plus the \$20 membership fee.



What information from the context led you to know that variables would be on both sides of the inequality?

Situation C

Ihab is training for a marathon. He runs at a pace of 5 miles per hour (mph). In the past week, the distances of his runs have ranged from 4.5 miles to 20 miles. He writes an inequality that can be used to find the range of time he's spent running in the past week.

Matching Inequality and Solution

$$\begin{aligned} 4.5 &\leq 5k \leq 20 \\ 0.9 &\leq k \leq 4 \end{aligned}$$

How does the inequality match the situation?

Answers will vary. Ihab runs at a pace of 5 miles per hour, k , with a range with a low of 4.5 miles and a high of 20 miles. This inequality accounts for a run in the middle, knowing the value is somewhere between 4.5 and 20.



How can you tell from the context that the resulting inequality would be a compound inequality?

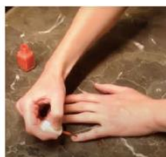
6.3.3 Exploration: Is There Enough Time?

Component Overview: Students will complete a three-act task with timing and painting nails, connecting the concept to inequalities.



6.3.3 Exploration: Is There Enough Time?

1. Candace is getting ready to leave for the movies. She decides at the last minute to try to paint her nails before she leaves.



It's 6:40 p.m., and Candace needs to leave at 7:00 p.m. What information do you need to determine if she will have enough time to paint her nails?

Answers will vary. How long it takes to paint her nails. How long it takes to paint each nail. How long it takes for the nail polish to dry.

2. The images shown give additional information about the situation. The first image shows it takes her 42 seconds to paint the first three nails. The second image gives the amount of time she needs to let her nails dry before leaving.



- a. How long does it take her to paint each nail, on average?
14 seconds
- b. Explain whether or not you think Candace will have enough time to paint her nails.
Answers will vary. Yes, she does have enough time because it will take her a little longer than 17 minutes in total to paint her nails and have them dry. She has 20 minutes until she needs to leave.



Don't let the space on the page fool you, this activity is meant to be a quick lead-in to the 6.3.4 Guided Instruction.

To facilitate this activity, the following videos will be needed which can be found in the digital area. The paper-based component is set up to be able to be completed without the videos if it is not possible to play them.

- Act 1: (Question 1)
- Act 2: (Question 2)
- Act 3: (After Question 2)

6.3.4 Guided Instruction: Writing Inequalities to Represent Real-World Contexts

Component Overview: Students will learn how to write inequalities to represent a real-world context and interpret the viability of solutions.



6.3.4 Guided Instruction: Writing Inequalities to Represent Real-World Contexts

1. The nail polish scenario provides an example of how inequalities can be used to solve problems in the real world.
 - a. Let t be the time it takes to paint one nail, in minutes. Write an expression in terms of t that represents the total amount of time it takes for the nail polish process to be complete, including 15 minutes for drying.
 $10t + 15$
 - b. In the Exploration, Candace had a maximum of 20 minutes. Using your expression from Part A, write an inequality that represents this situation.
 $10t + 15 \leq 20$
 - c. Solve the inequality.
 $10t + 15 \leq 20$
 $10t \leq 5$
 $t \leq \frac{5}{10}$
 $t \leq \frac{1}{2}$
 - d. Explain how the maximum value in the solution set relates to the nail polish situation.
It represents the number of minutes that can be spent painting each nail to finish in time.
 - e. Graph the solution set.



This activity teaches students how to write inequalities to represent real-world contexts using the nail polish scenario from the previous component. Each sub-question of this activity is a scaffolded approach to building understanding of how to write, solve, and interpret the solution to an inequality representing a real-world scenario.



- What does a fraction mean in this situation?
- What does $\frac{1}{2}$ mean in terms of minutes and seconds?
- What interval makes sense for this situation?

6.3.4 Guided Instruction: Writing Inequalities to Represent Real-World Contexts (continued)

- f. The value -0.5 is in the solution set of the inequality. Discuss with your partner why this is not a viable solution in the real-world context. Write your explanation.

Time cannot be negative.

- g. Explain whether or not the value 0 is a viable solution in this context.

No, because it would mean she spent no time per nail.

- h. Select any statements that apply for values to be viable solutions in this context.

- ☐ Values must be non-negative.
- ☐ Values must be greater than 0.
- ☐ Values must be integers.
- ☐ Values cannot be more than ____ decimal places.
- ☐ Values cannot be greater than 0.5.

- i. Complete the compound inequality that represents the full set of possible solutions in the real-world context.

$$0 < t \leq \frac{1}{2}$$



For questions 1f and 1g, give students the opportunity to first discuss with a partner before completing as a whole group to ensure accuracy in responses, and provide students with a model to use when working through problems on their own.



Guide students through question 1h using a think aloud to model the type of thinking used when interpreting the viability of solutions in context. This is often where students get confused over minor misconceptions with real-world context and viable solutions.



Why is the $<$ symbol used when comparing t with 0 but the $\leq$ is used when comparing t with $\frac{1}{2}$?



Consider asking students to draw a representation of the real-world solution set on a number line.

Ask how this would compare to the graph of the mathematical solution set.

6.3.5 Activity: Gallery Walk

Component Overview: Students will engage in a Gallery Walk that consists of four scenarios where they will write the inequality, find the solution, create the graph, and determine viable solutions in the set. For this activity, students should work in small groups or with a partner.

6.3.5 Activity: Gallery Walk

Work with your partner or small group to complete the following for each real-world context.

- Write an inequality to represent the situation, including defining the variable used.
- Solve the inequality and graph the solution.
- Identify and explain any non-viable solutions.

Context A – Vacation Savings	
Inequality	$600 \leq 15h + 97.5 \leq 825$
Solution	$33.5 \leq h \leq 48.5$
Graph	
Non-Viable Solutions in the Set	Check any that apply. ☐ Solutions must be non-negative ☐ Solutions must be integers Explain. <i>Answers will vary. Because the values must be between 33.5 and 48.5, the first condition applies to any value that satisfies the inequality. However, it is possible for solutions to be non-integers because the problem context states she can work fractions of hours.</i>



If pacing becomes a concern, one of the contexts could be removed, leaving 3 contexts for the Gallery Walk. An idea would be to make two copies of each and have students rotate in two smaller groups, or all four could be left for them to decide how they want to approach it. If you choose to remove a context, consider removing D.



Consider modifying activities to provide additional scaffolds, if needed. Consider providing one of the answer choices for students who need the additional support and arrange the groups in such a way that only the students who need the additional support receive it.

6.3.5 Activity: Gallery Walk (continued)

Context B – Non-Operating Temperatures	
Inequality	$-25 \leq \frac{5}{9}(x - 32) \leq 45$
Solution	$-13^{\circ}\text{F} \leq x \leq 113^{\circ}\text{F}$
Graph	
Non-Viable Solutions in the Set	Check any that apply. ☐ Solutions must be non-negative ☐ Solutions must be integers Explain. <i>Answers will vary. In this context, negative temperatures are allowed. Since it is temperatures, they can be measured with fractional values.</i>

Context C – Gym Memberships	
Inequality	$8x + 15 < 6.75x + 22$
Solution	$x < 5.6$ $x \leq 5$ because of context, you can't take a partial class
Graph	
Non-Viable Solutions in the Set	Check any that apply. ☐ Solutions must be non-negative ☐ Solutions must be integers Explain. <i>Answers will vary. In this real-world context, a person cannot take partial nor negative classes.</i>



Would a single inequality or a compound inequality better represent this situation? Why?



How can you determine the viability of solutions for a specific real-world context?

6.3.5 Activity: Gallery Walk (continued)

Context D – Planting Citrus Trees	
Inequality	$60 \leq 4s + 15.2 \leq 100$
Solution	$11.2 \leq s \leq 21.2$
Graph	
Non-Viable Solutions in the Set	Check any that apply. ☐ Solutions must be non-negative ☐ Solutions must be integers Explain. <i>Answers will vary. The answers must be between 11.2 and 21.2, but this can include non-integer values.</i>



At the conclusion of this activity, consider engaging in a whole-group discussion about strategies students used to determine the answers to each of these real-world contexts.

6.3.6 Wrap-Up: Reflection

Component Overview: Students will complete a Growth Mindset reflection activity. Students should complete this independently.



6.3.6 Wrap-Up: Reflection

Answers will vary.

Which mindset did I demonstrate?	Choose the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to “un-stick” myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I re-read the problem to ensure understanding of the context?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I put as much effort as I possibly could into the problems?	☐ Never ☐ Sometimes ☐ Usually ☐ Always



Consider pairing students to discuss their reflections and share strategies that helped them when they were stuck.